SEMA 4.x EC Linux GPIO interrupt handling guidelines

Supported Hardware

1. Express-TL
2. cExpress-TL
3. Express-RLP
4. cExpress-RLP
5. cExpress-MTL

Note: cExpress-MTL works only with Linux kernel 6.0 and above.

Supported Operating Systems

The EC GPIO interrupt driver has been tested on the following operating systems:

1. Ubuntu 20, 22, 24
2. Redhat 9.5
3. Cent OS 10
4. Debian 12
5. SUSE Linux 15-SP6
6. Fedora 41

Loading the EC GPIO interrupt driver

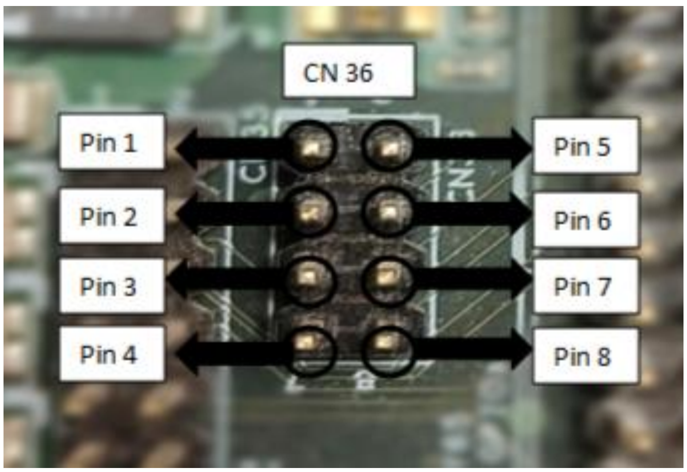
1. Extract the “sema-ec-linux-irq.zip” file containing the driver source.
2. Open the terminal in the directory containing the driver source.
3. In the terminal enter “**sudo make install_irq**” to compile and install the driver.
4. Next, we need to load the driver and pass the trigger type as an argument.
    a. For edge triggers “**sudo modprobe adl-ec-gpio-irq trigger_type=edge**”.
    b. For low-level triggers “**sudo modprobe adl-ec-gpio-irq trigger_type=low-level**”.
    c. For high-level triggers “**sudo modprobe adl-ec-gpio-irq trigger_type=high-level**”.

5. To change the trigger type after loading the driver, unload the driver by running the command “**sudo modprobe -r adl-ec-gpio-irq**” and reload the driver with the required trigger type as the argument.

Testing the GPIO interrupts using SEMA

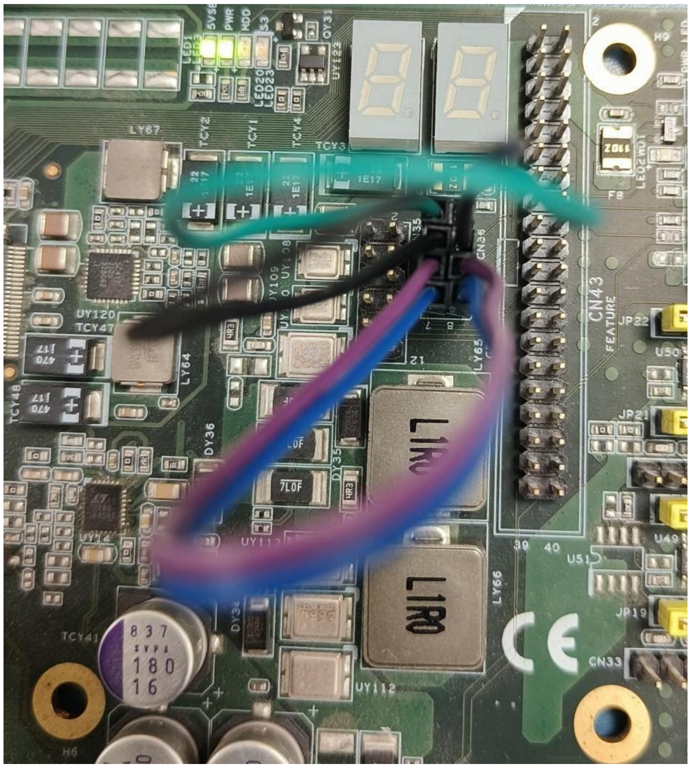
Set up the hardware

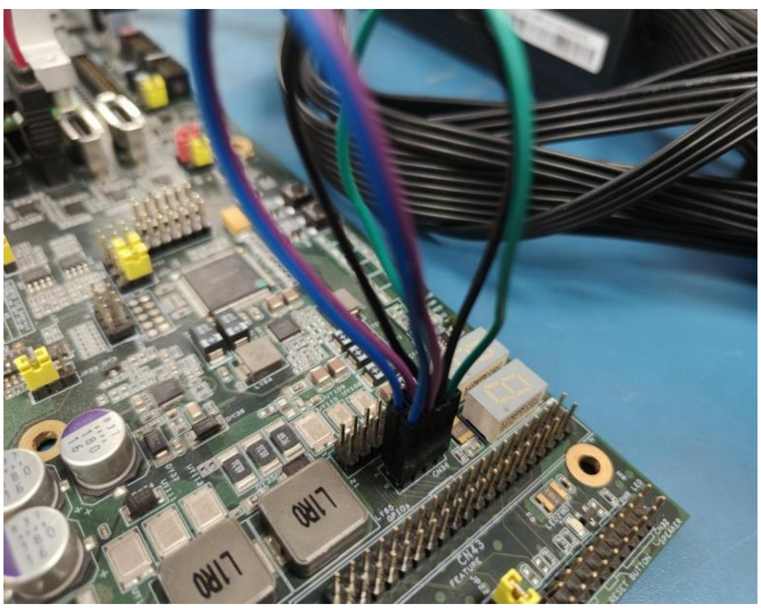
1. In the carrier board there are four GPI (Input) pins and four GPO (output) pins in CN36.



2. In CN36 (Default state)
   Pin 1 – GPI0 (INPUT)
   Pin 2 – GPI1 (INPUT)
   Pin 3 – GPI2 (INPUT)
   Pin 4 – GPI3 (INPUT)
   Pin 5 – GPO0 (OUTPUT)
   Pin 6 – GPO1 (OUTPUT)
   Pin 7 – GPO2 (OUTPUT)
   Pin 8 – GPO3 (OUTPUT)

3. Connect GPI0 to GPO0, GPI1 to GPO1, GPI2 to GPO2 and GPI3 to GPO3.





4. Pins 1 to 4 are GPI pins and pins 5 to 8 are GPO pins.

Load the SEMA drivers

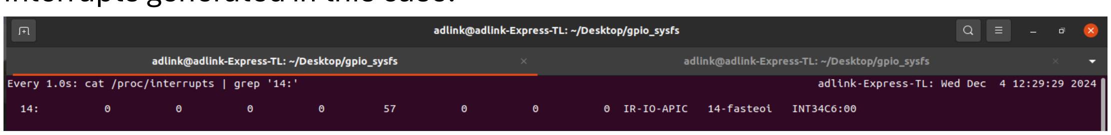
1. Extract the zip file “sema-ec-linux-irq.zip” containing the SEMA source.
2. Open the terminal in the directory containing the SEMA source and enter the command “**sudo make install**” to compile and install the SEMA drivers.
3. To load the SEMA drivers, enter the command: “**sudo modprobe -a adl-ec adl-ec-boardinfo adl-ec-vm adl-ec-wdt adl-ec-hwmon adl-ec-nvmem adl-ec-bklight adl-ec-i2c adl-ec-gpio adl-ec-nvmem-sec**”

Testing the EC GPIO interrupts

Edge Interrupts

To test the Edge interrupts:

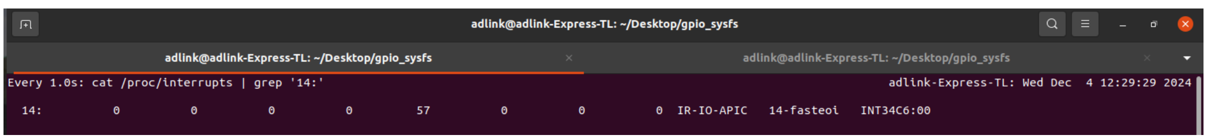
1. Change the state of any one of the output pins to high and low continuously, you can do this in two ways:
   i. Run the shell script “irq.sh” by running the command “**. irq.sh**”. This shell script will continuously change the state of Pin 8 (GPO3) to high and low every one second and this will generate Edge triggered interrupts.
   ii. You can also change the pin state by running the semautil commands, for example, to change the GPO0 pin to high from low enter the command “**sudo semautil /g set_level 5 1**”, to chan
   iii. ge the GPO0 pin to low from high enter the command “**sudo semautil /g set_level 5 0**”.
2. To observe the total number of interrupts generated, enter the command “**sudo watch -n 1 “cat /proc/interrupts | grep ‘14:’””**, and you will get the output like shown in the image below. The number 57 is the total number of GPIO interrupts generated in this case.



Low-level interrupts

To test the Low-level interrupts:

1. All the GPO pins must be set to high initially. To do so you can run the following commands.
    a. "**sudo semautil /g set_level 5 1**" to set GPO0 high.
    b. "**sudo semautil /g set_level 6 1**" to set GPO1 high.
    c. "**sudo semautil /g set_level 7 1**" to set GPO2 high.
    d. "**sudo semautil /g set_level 8 1**" to set GPO3 high.
2. Next to trigger a low-level interrupt, change the level of any one of the GPO pins to a low level. For example, to change the GPO1 (pin 6) to low, enter the command "**sudo semautil /g set_level 6 0**".
3. Low-level interrupts will be continuously generated till the level of the pin made low in the step above is made high. For example, "**sudo semautil /g set_level 6 1**".
4. To observe the total number of interrupts generated, enter the command "**sudo watch -n 1 "cat /proc/interrupts | grep '14:'"**", and you will get the output like shown in the image below. The number 57 is the total number of GPIO interrupts generated in this case.

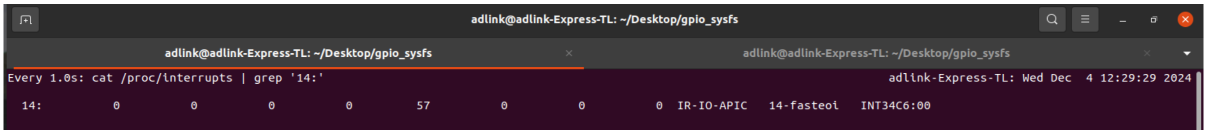


High-level interrupts

To test the High-level interrupts:

1. All the GPO pins must be set to low initially. To do so you can run the following commands.
    a. "**sudo semautil /g set_level 5 0**" to set GPO0 low.
    b. "**sudo semautil /g set_level 6 0**" to set GPO1 low.
    c. "**sudo semautil /g set_level 7 0**" to set GPO2 low.
    d. "**sudo semautil /g set_level 8 0**" to set GPO3 low.
2. Next to trigger a high-level interrupt, change the level of any one of the GPO pins to a high level. For example, to change the GPO2 (pin 7) to high, enter the command "**sudo semautil /g set_level 7 1**".
3. High level interrupts will be continuously generated till the level of the pin made high in the step above is made low. For example, "**sudo semautil /g set_level 7 0**".

4. To observe the total number of interrupts generated, enter the command "**sudo watch -n 1 "cat /proc/interrupts | grep '14:'"**", and you will get the output like shown in the image below. The number 57 is the total number of GPIO interrupts generated in this case.



<u>Note</u>: Please do not copy and paste the commands in the Linux terminal. Please type and enter the commands in the Linux terminal.